

Regression with dummy variables and categorical features

07

Outline

- (30) Introduction: Two levels → Dummy variable
- (15) Multi-level dummy variables
- (15) Seasonal dummy variables
- (15) Applications
- (10) Interaction

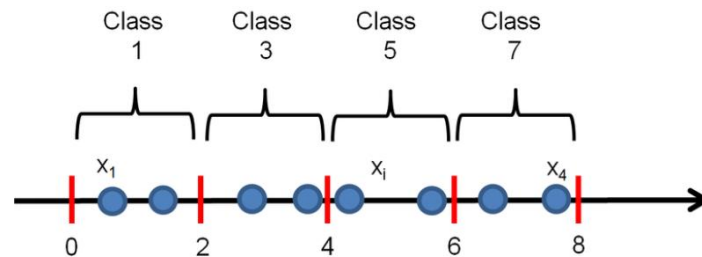
review: type of data

Numerical

- A variable is *numerical* if meaningful arithmetic can be performed on it.
- A numerical variable is *discrete* if it results from a count, such as the number of children.
- A *continuous* variable is the result of an essentially continuous measurement, such as weight or height.

Categorical/Factor

- Examples
 - Order fulfillment: on-time vs delayed
 - Shipment type: domestic vs international
 - Route type: direct vs transshipment
 - Mode of transport: air, sea, road, rail
 - Hotel customer segment: Online travel agent (OTA), traditional travel agent, Walk-in, Corporate
 - Days of week: Mon, Tue, ..., Fri, Sat, Sun
- A categorical variable is *ordinal* if there is a natural ordering of its possible values. If there is no natural ordering, it is *nominal*.
- A *dummy* variable is a 0–1 coded variable for a specific category.
 - It is coded as 1 for all observations in that category and 0 for all observations not in that category.



Categorizing a numerical variable as categorical is called *binning* (putting the data into discrete bins) or *discretizing*.
Examples: aging inventory, kid/adult/senior



Example: Bank Salary

N: numerical

F: factor

	N	F	F	N	N	F	F	N	N
Employee	EducLev	JobGrade	Age	YrsPrior	PCJob	Gender	YrsExper	Salary	
1	3	1	26	1	No	Male	3	\$32,000	
2	1	1	38	1	No	Female	14	\$39,100	
3	1	1	35	0	No	Female	12	\$33,200	
4	2	1	40	7	No	Female	8	\$30,600	
5	3	1	28	0	No	Male	3	\$29,000	
...									
208	5	6	62	0	No	Female	33	\$30,000	

- EducLev: education level
 - 1 finished high school
 - 2 finished some college courses
 - 3 obtained bachelor's degree
 - 4 took some graduate courses
 - 5 obtained graduate degree
- JobGrade: current job level, the possible levels being 1 through 6 (highest)
- Age: employee's current age
- YrsPrior: # of years of work experience at another bank prior to working at this bank
- PCJob: whether the employee's current job is computer-related
- Gender
- YrsExper: years of experience with this bank
- Salary: current annual salary

Example: Bank Salary

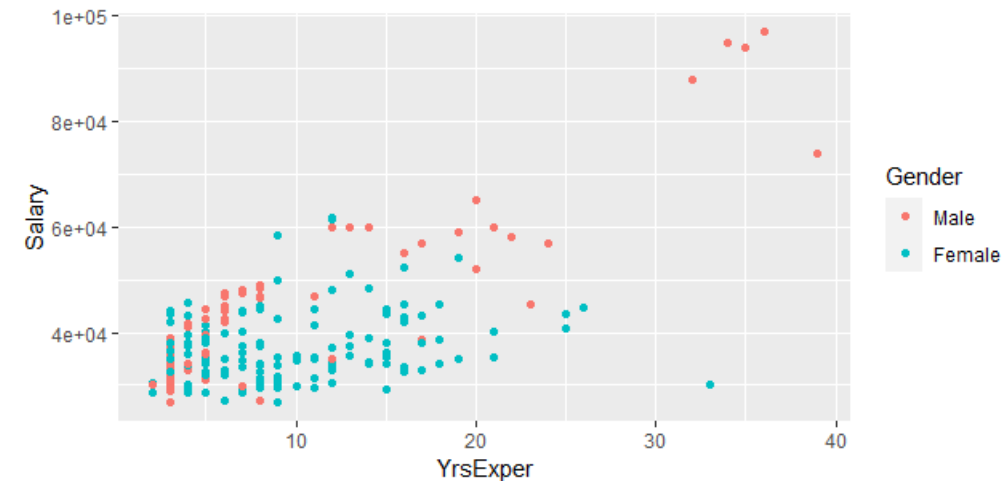
The Fifth National Bank of Springfield is facing a gender discrimination suit that its female employees receive smaller salaries.

1. Naïve approach: Compare the avg female salary to the avg male salary
2. Hypothesis test: Which one (**Paired t -test** or **Two-sample t -test**)?
3. Regression: The question is whether the difference between female and male salaries is still evident after taking the other attributes (e.g., working experiences) into account.

Row Labels	Average of Salary
Female	37,210
Male	45,505
Grand Total	39,922

```
d %>%
  group_by(Gender) %>%
  summarise(mean = mean(Salary))
```

```
d %>%
  ggplot() +
  geom_point(mapping = aes(x=YrsExper, y=Salary, colour=Gender))
```



Example: Bank Salary

- Dependent variable

$Y = \text{Salary}$

- Independent variables

1. $X = \text{YrsExper}$

2. $1\{F\} = 1\{\text{observation is Female}\} = \begin{cases} 1, & \text{if female} \\ 0, & \text{otherwise} \end{cases}$
Dummy variable

- Regression equation

$$\hat{Y} = \beta_0 + \beta_1 X + \beta_2 1\{F\}$$

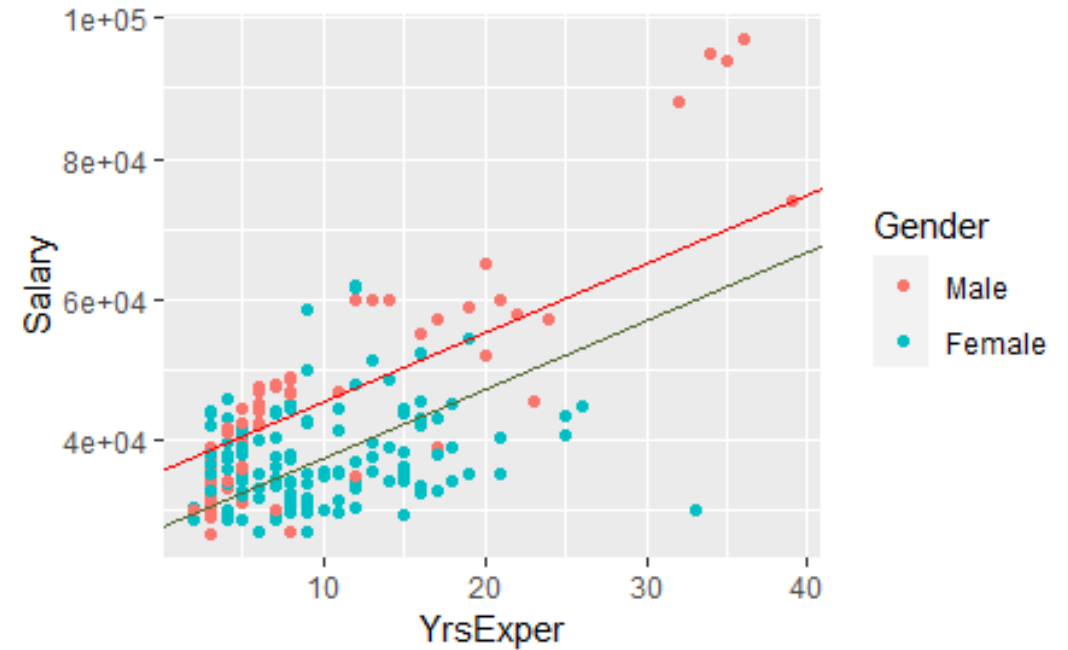
$$\hat{Y} = 35823.8 + 981.15X - 8011.89 1\{F\}$$

- Regression for male ($F = 0$)

$$\hat{Y} = \beta_0 + \beta_1 X = 35823.8 + 981.2 X$$

- Regression for female ($F = 1$)

$$\hat{Y} = (\beta_0 + \beta_2) + \beta_1 X = 27811.9 + 981.2 X$$



Coefficient interpretations

- Intercept $\beta_0 = 35823.8$
 - For male employees (since $F = 0$), the predicted salary with $\text{YrsExper} = 0$ is 35823.80 USD
 - This represents the starting salary for males (baseline group)
- Slope for YrsExper $\beta_1 = 981.15$
 - Each additional year of experience increases salary by 981.15 USD, holding gender constant.
 - This effect applies equally for males and females (parallel lines in graph)
- Gender dummy $\beta_2 = -8011.89$
 - For the same years of experience, females earn 8011.89 less than males on average.
 - The coefficient is negative, showing the downward shift in the regression line for females.



Example: Bank Salary

- Dependent variable

$Y = \text{Salary}$

- Independent variables

1. $X = \text{YrsExper}$

2. $1\{F\} = 1\{\text{observation is Female}\} = \begin{cases} 1, & \text{if female} \\ 0, & \text{if male (baseline)} \end{cases}$
Dummy variable

- Regression equation

$$\hat{Y} = \beta_0 + \beta_1 X + \beta_2 1\{F\}$$

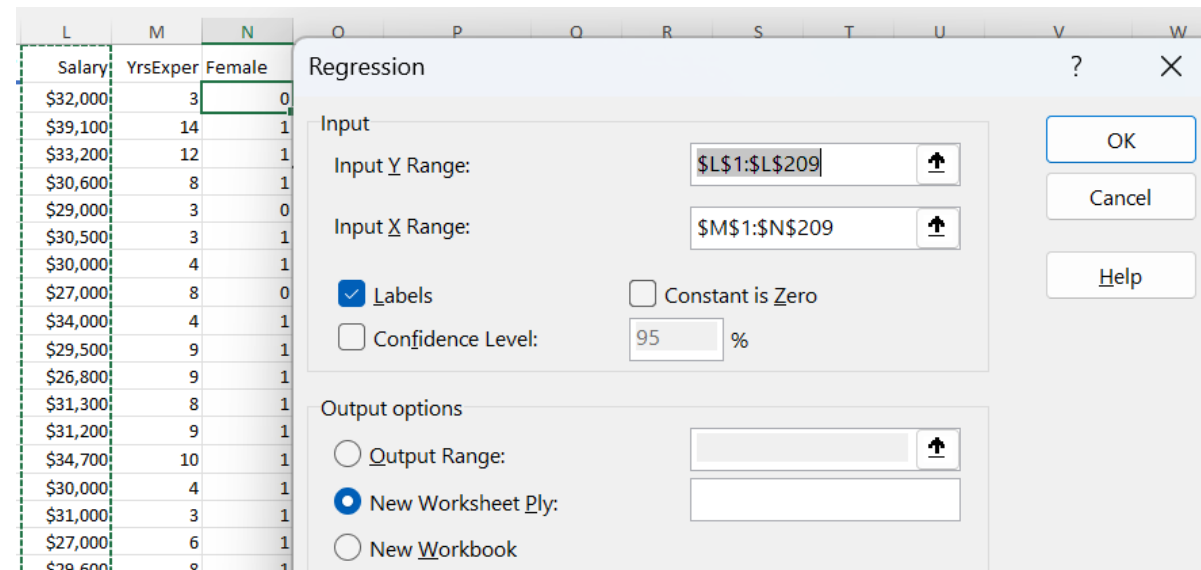
$$\hat{Y} = 35823.8 + 981.15X - 8011.89 1\{F\}$$

- Regression for male ($F = 0$)

$$\hat{Y} = \beta_0 + \beta_1 X = 35823.8 + 981.2 X$$

- Regression for female ($F = 1$)

$$\hat{Y} = (\beta_0 + \beta_2) + \beta_1 X = 27811.9 + 981.2 X$$



Sheet: salaryAll

Use of indicator	Salary	YrsExper	Female		Coefficients
	\$32,000	3	0	Intercept	35823.79
	\$39,100	14	1	YrsExper	981.15
	\$33,200	12	1	Female	-8011.89
	\$30,600	8	1		
	\$29,000	3	0		
	\$30,500	3	1		
	\$30,000	4	1		
	\$27,000	8	0		
	\$34,000	4	1		
	\$29,500	9	1		



R: Factor vs Numerical columns

CSV File: banksalary

Excel File 09-Regression-Dummy Sheet banksalary

n	f	f	n	n	f	f	n	n
Employee	EducLev	JobGrade	Age	YrsPrior	PCJob	Gender	YrsExper	Salary
1	3	1	26	1	No	Male	3	\$32,000
2	1	1	38	1	No	Female	14	\$39,100
...								
208	5	6	62	0	No	Female	33	\$30,000

- factor/categorical: f
- Numerical: n

```
d <- read_csv("banksalary.csv", col_types = "nffnffnn")
glimpse(d)
```

```
Rows: 208
Columns: 9
$ Employee <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, ~
$ EducLev  <fct> 3, 1, 1, 2, 3, 3, 3, 3, 1, 3, 3, 2, 2, 2, 3, 3, 2, 3, 2, 1~
$ JobGrade <fct> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ Age      <dbl> 26, 38, 35, 40, 28, 24, 27, 33, 62, 31, 34, 37, 37, 58, 33, 27, 30,~
$ YrsPrior <dbl> 1, 1, 0, 7, 0, 0, 0, 2, 0, 0, 2, 8, 0, 6, 0, 0, 0, 9, 6, 3, 2, 3, 0~
$ PCJob    <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No, No, No, No,~
$ Gender   <fct> Male, Female, Female, Female, Male, Female, Female, Male, Female, F~
$ YrsExper <dbl> 3, 14, 12, 8, 3, 3, 4, 8, 4, 9, 9, 8, 9, 10, 4, 3, 6, 8, 5, 4, 4, 3~
$ Salary   <dbl> 32000, 39100, 33200, 30600, 29000, 30500, 30000, 27000, 34000, 2950~
```

```
de <- read_excel("09-Regression-Dummy.xlsx", sheet = "banksalary")
de$EducLev <- as.factor(de$EducLev)
glimpse(de)
class(de$EducLev)
```


Coefficient interpretation

Quick check

Dependent variable:

$Y = \text{Salary}$

Independent variables:

1. $X_1 = \text{YrsExper}$ — years of experience (continuous)
2. $1\{\text{PCJob}\}$ — dummy variable:

$$1\{\text{PCJob}\} = \begin{cases} 1 & \text{if the employee uses a PC in the job} \\ 0 & \text{otherwise} \end{cases}$$

2. Regression equation

$$\hat{Y} = \beta_0 + \beta_1 \text{YrsExper} + \beta_2 1\{\text{PCJob}\}$$

Example (you can adjust once we estimate from the CSV):

$$\hat{Y} = 33,500 + 950 \text{YrsExper} + 6,200 1\{\text{PCJob}\}$$

It does not matter which of the two variables is used as a baseline

Write regression equations for the two cases
PCJob = 0 (baseline)

PCJob = 1

Coefficient interpretations

- $\beta_1 = 950$
- $\beta_2 = 6200$

Dummy variables

From binary to multi-level dummy variables



Indicator (Dummy) Variables

- When observations fall into one of two categories, we distinguish them by means of an **indicator** or **dummy** variable, defined as

$$1\{A\} = \begin{cases} 1, & \text{if the observation falls into category A} \\ 0, & \text{if the observation falls into category B} \end{cases}$$

- In general, when there are m **categories (contrasts)**, let

$$D_i = 1\{\text{category } i\} = \begin{cases} 1, & \text{if the observation falls into category } i \\ 0, & \text{otherwise} \end{cases}$$

Bank Example: EducLevel

EducLevel has $m = 5$ categories.

Let $E_i = 1\{\text{if EducLev} = \text{category } i\}$ for $i = 1, 2, \dots, 5$

EducLev		Edu1 E_1	Edu2 E_2	Edu3 E_3	Edu4 E_4	Edu5 E_5
1	finished high school	1	0	0	0	0
2	finished some college courses	0	1	0	0	0
3	obtained bachelor's degree (baseline)	0	0	1	0	0
4	took some graduate courses	0	0	0	1	0
5	obtained graduate degree	0	0	0	0	1

Indicator (Dummy) Variables

- In general, when there are m categories (also called **contrasts** or **levels**), we need only $(m - 1)$ dummy variables, which can be defined as follows for $i = 2, 3, \dots, m$:

$$D_i = 1\{\text{category } i\} = \begin{cases} 1, & \text{if the observation falls into category } i \\ 0, & \text{otherwise} \end{cases}$$

- All indicator variables have the value 0 for category i (**baseline**)
- Coefficient of dummy variable for a particular category measures the differential effect of being in that category, compared to the baseline (for which all dummies equal 0).

Bank Example: EducLevel

- EducLevel has $m = 5$ categories.
Let $E_i = 1\{\text{if EducLev} = \text{category } i\}$ for $i = 1, 2, \dots, 5$
- We need $5 - 1 = 4$ dummy variables.
- Let the **baseline** be E_3 .
Dummy variables needed are E_1, E_2, E_4, E_5

E_3 is not independent of E_1, E_2, E_4, E_5 .

$$E_1, E_2, E_4, E_5 \leftrightarrow E_3$$

e.g.,

- $E_1 = 1, E_2 = 0, E_4 = 0, E_5 = 0 \leftrightarrow E_3 = 0$
- $E_3 = 1 \leftrightarrow E_1 = 0, E_2 = 0, E_4 = 0, E_5 = 0$. Note that all dummy variables are zero for the baseline, Edu3

		Edu1	Edu2	Edu3	Edu4	Edu5
EducLev		E_1	E_2	E_3	E_4	E_5
1	finished high school	1	0	0	0	0
2	finished some college courses	0	1	0	0	0
3	obtained bachelor's degree (baseline)	0	0	1	0	0
4	took some graduate courses	0	0	0	1	0
5	obtained graduate degree	0	0	0	0	1



Indicator (Dummy) Variables

Employee	Salary	YrsExper	YrsPrior	Female	EducLev
1	\$32,000				3
2	\$39,100				1
3	\$33,200				1
...					
208	\$30,000				5

Factor column
with 5 categories

EducLev: education level

- 1 finished high school
- 2 finished some college courses
- 3 obtained bachelor's degree
- 4 took some graduate courses
- 5 obtained graduate degree



Employee	Salary	YrsExper	YrsPrior	Female	Edu1	Edu2	Edu4	Edu5
1	\$32,000				0	0	0	0
2	\$39,100				1	0	0	0
3	\$33,200				1	0	0	0
...								
208	\$30,000							

4 dummy variables
with baseline = Edu3





Example: Bank Salary with Education level

- Independent variables

- $X = \text{YrsExper}$
 - $P = \text{YrsPrior}$
 - $1\{F\} = 1\{\text{observation is Female}\}$
 - $E_i = 1\{\text{if EducLev} = \text{category } i\} \text{ for } i = 1, 2, 4, 5$
- Note that the baseline E_3 is not included.

- Regression equation

$$\hat{Y} = \beta_0 + \beta_1 X + \beta_2 P + \beta_3 1\{F\} + \gamma_1 E_1 + \gamma_2 E_2 + \gamma_4 E_4 + \gamma_5 E_5$$

Predicted Salary = 31377.9 + 1032.69YrsExper + 362.2YrsPrior -4501.3 1{Gender=Female}
 -4764.6 1{Edu=1} -4604.3 1{Edu=2} + 2555.3 1{Edu=4} + 7005.3 1{Edu=5}

Salary	YrsExper	YrsPrior	Female	Edu1	Edu2	Edu4	Edu5		Coefficients
\$32,000	3	1	0	0	0	0	0	Intercept	31377.91
\$39,100	14	1	1	1	0	0	0	YrsExper	1032.93
\$33,200	12	0	1	1	0	0	0	YrsPrior	362.23
\$30,600	8	7	1	0	1	0	0	Female	-4501.32
\$29,000	3	0	0	0	0	0	0	Edu1	-4764.56
\$30,500	3	0	1	0	0	0	0	Edu2	-4604.31
\$30,000	4	0	1	0	0	0	0	Edu4	2555.28
\$27,000	8	2	0	0	0	0	0	Edu5	7005.65
\$34,000	4	0	1	1	0	0	0		
\$29,500	9	0	1	0	0	0	0		
\$26,800	9	2	1	0	0	0	0		

```
d <- read_csv("banksalary.csv", col_types = "nffnnffnn")
```

```
m3 <- lm(Salary ~ YrsExper + YrsPrior + Gender + EducLev, d)
summary(m3)
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	31377.9	1340.6	23.405	< 2e-16	***
YrsExper	1032.9	69.6	14.840	< 2e-16	***
YrsPrior	362.2	158.1	2.291	0.02302	*
GenderFemale	-4501.3	1085.8	-4.146	5.00e-05	***
EducLev1	-4764.6	1473.4	-3.234	0.00143	**
EducLev2	-4604.3	1489.5	-3.091	0.00228	**
EducLev5	7005.6	1232.3	5.685	4.58e-08	***
EducLev4	2555.3	2563.8	0.997	0.32013	

Multiple R-squared: 0.6449, Adjusted R-squared: 0.6324



Example: Bank Salary Education level

Regression equation

Predicted Salary = **31377.9** + 1032.69YrsExper + 362.2YrsPrior **-4501.3** 1{Gender=Female}
-4764.6 1{Edu=1} -4604.3 1{Edu=2} + 2555.3 1{Edu=4} + **7005.3** 1{Edu=5}

- Females at the 5th education level:

Predicted Salary = **33882.24** + 1032.69YrsExper + 362.2YrsPrior

Intercept **31377.9 – 4501.3 + 7005.3 = 33882**

```
> as.numeric(m3$coefficients[1] + m3$coefficients["GenderFemale"]  
+ m3$coefficients["EducLev5"])  
[1] 33882.24
```

- Male at the 2nd education level:
- We can write a separate equation for *any combination* of categories by setting dummies to appropriate values.
How many equations?



Example: Bank Salary Education level

Regression equation

$$\text{Predicted Salary} = 31377.9 + 1033 \text{ YrsExper} + 362 \text{ YrsPrior} - 4501 \text{ 1}\{\text{Gender=Female}\} \\ - 4765 \text{ 1}\{\text{Edu=1}\} - 4604 \text{ 1}\{\text{Edu=2}\} + 2555 \text{ 1}\{\text{Edu=4}\} + 7005 \text{ 1}\{\text{Edu=5}\}$$

Coefficient interpretation

- For either gender and any education level,
 - the expected increase in salary for one extra year of experience with the bank is 1033 USD;
 - the expected increase in salary for one extra year of experience with another bank is 362 USD.
- Note that the **baseline** corresponds to Edu = 3 bachelor's degree.
 - An employee with education level 5 can expect to earn 7005 USD more than an employee with education level 3, all else being equal.
 - An employee with education level 1
- The coefficient -4501 USD for female indicates the average salary disadvantage for females relative to male, given that they have the same experience level and the same education level.

Mult-level dummy variables

Quick check: Creating dummy variables for multi-level categories

- Recall JobGrade: current job level, the possible levels being 1 through 6 (highest)
- Baseline: Level 1
- How many dummy variables are needed?
- Specify these multi-level dummy variables. Let $G_i = 1\{\text{JobGrade}=i\}$

JobGrade	G_1	G_2	G_3	G_4	G_5	G_6
1						
2						
3						
4						
5						
6						

Baseline: Level 1

$$\hat{Y} = \beta_0 + \beta_1 X + \beta_2 P + \beta_3 1\{F\} + \gamma_2 G_2 + \gamma_3 G_3 + \gamma_4 G_4 + \gamma_5 G_5 + \gamma_6 G_6$$

In-class. Write forecast equation for the following cases

- Female JobGrade2 $\hat{Y} = (\beta_0 + \beta_3 + \gamma_2) + \beta_1 X + \beta_2 P$
- Female JobGrade1
- Male JobGrade3

Seasonal dummy variables



Seasonal Dummy Variables

Seasonal indicators

- Monthly data: 11 dummy variables
- Quarterly data: ?
- Daily data: Day of the week (DoW) as predictor

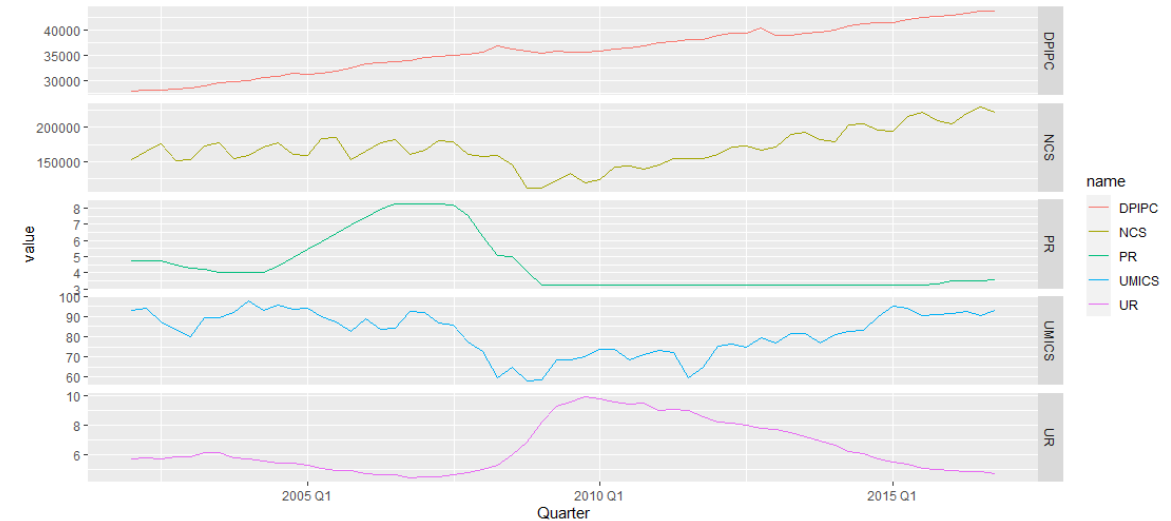
t		$d_{1,t}$	$d_{2,t}$	$d_{3,t}$	$d_{4,t}$	$d_{5,t}$	$d_{6,t}$
1	Monday	1	0	0	0	0	0
2	Tuesday	0	1	0	0	0	0
3	Wednesday	0	0	1	0	0	0
4	Thursday	0	0	0	1	0	0
5	Friday	0	0	0	0	1	0
6	Saturday	0	0	0	0	0	1
7	Sunday	0	0	0	0	0	0
8	Monday	1	0	0	0	0	0
$\vdots$		$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Let $d_{i,t} = 1$ if data in row t corresponds to DoW i ($i = 1$ Mon, $i = 2$ Tue, ...)

Baseline: Sunday (for which all the dummies equal 0 $d_{1,t} = d_{2,t} = \dots = d_{5,t} = d_{6,t} = 0$)

If we want baseline to be Monday, which dummy variables to be used?

Example New Car Sales (Con.)



- Model 1: $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$

$$\text{NCS.FC} = 25428.255 + 3.228\text{DPIPc} - 7666.852\text{UR} - 3219.499\text{PR} + 1122.429\text{UMICS}$$

- Model 2 (inclass): $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \gamma_2 Q_2 + \gamma_3 Q_3 + \gamma_4 Q_4$

$$\text{NCS.FC} = 15351.530 + 3.197\text{DPIPc} - 7607.513\text{UR} - 3306.902\text{PR} + 1154.704\text{UMICS} + 14346.704Q_1 + 18654.821Q_3 + 1287.332Q_4$$

- Model 3 (Homework): $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \gamma_2 Q_2 + \gamma_3 Q_3 + \gamma_4 Q_4 + \alpha t$

- Model 3 captures both **trend** and season, while Model 2 captures only season.

Model	R^2	Adjusted R^2
1	0.8282	0.8157
2	0.9177	0.9066
3 (Homework)		21



Significance of dummy variables

Do **NOT** drop “insignificant” seasonal (or education) dummies one-by-one.

- The exclusion of some seasonal dummies because their estimated coefficients are not significant is not recommended.
- Seasonality is a **single compound hypothesis**, so the F-test should be used to test the factor as **block**, instead of the t-test.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	26613.4	1794.1	14.833	< 2e-16 ***
YrsExper	1032.9	69.6	14.840	< 2e-16 ***
YrsPrior	362.2	158.1	2.291	0.02302 *
GenderFemale	-4501.3	1085.8	-4.146	5.00e-05 ***
Edu_2	160.2	1656.0	0.097	0.92301
Edu_3	4764.6	1473.4	3.234	0.00143 **
Edu_4	7319.8	2694.2	2.717	0.00717 **
Edu_5	11770.2	1510.2	7.794	3.49e-13 ***

Multiple R-squared: 0.6449, Adjusted R-squared: 0.6324

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	31377.9	1340.6	23.405	< 2e-16 ***
YrsExper	1032.9	69.6	14.840	< 2e-16 ***
YrsPrior	362.2	158.1	2.291	0.02302 *
GenderFemale	-4501.3	1085.8	-4.146	5.00e-05 ***
Edu_1	-4764.6	1473.4	-3.234	0.00143 **
Edu_2	-4604.3	1489.5	-3.091	0.00228 **
Edu_4	2555.3	2563.8	0.997	0.32013
Edu_5	7005.6	1232.3	5.685	4.58e-08 ***

Multiple R-squared: 0.6449, Adjusted R-squared: 0.6324

Note that the coefficient of 1{Edu=4} is not significant. Should we drop the nonsignificant values? NO!

Test the factor as a block

$H_0 : \gamma_2 = \dots = \gamma_m = 0$ (no factor effect)

Use nested models:

Reduced: without the factor

Full: with the factor

$$F = \frac{(RSS_{red} - RSS_{full}) / (p_{full} - p_{red})}{RSS_{full} / (n - p_{full})}$$

Reject $H_0 \Rightarrow$ keep the whole factor; otherwise, drop the whole factor (not one dummy at a time).

```
anova(lm(Salary ~ YrsExper + YrsPrior + Gender, d),
      lm(Salary ~ YrsExper + YrsPrior + Gender + EducLev, d))
```

These two models are equivalent.

Applications

Event modeling

Intervention

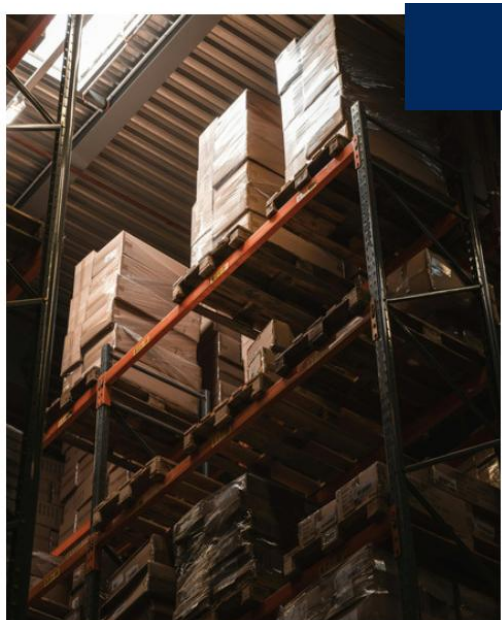
Applications of Dummy Variables

- Seasonality
- When forecasting daily sales, you want to take account of whether the day is a public holiday or not.
- Chinese New Year differs from most holidays because it is not held on the same date each year, and its effect can last for several days.
 - Dummy variable can be used with value 1, where holiday falls in the particular time period and 0 otherwise.
- Special events: “**event modeling**”
 - Promotion campaign (next example)
 - When forecasting a tourist arrivals to Brazil, we want to account for the effect of the Rio de Janeiro summer Olympics in 2016.
 - COVID outbreaks and government restrictions
- A dummy variable can be used to account for an **outlier** in the data.
- Intervention analysis

OPTIMIZING PERIODIC INVENTORY POLICY

CASE STUDY OF E-COMMERCE RETAILERS

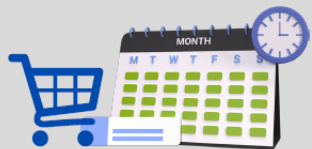
6620427003 WISARUTA CHAOLAKA



1 DATA COLLECTION & TRAIN/TEST SPLIT

FLAG CAMPAIGN (FACTOR TYPE)

- BAU** Normal day (Non Campaign)
- MidMonth** Campaign on the 15th.
- Payday** Campaign on the 25th.
- DDay** Campaign on Double-Digit days (e.g., 9,9, 11,11).
- BD** Brand-specific campaign day.



DATA PREPARATION

```
#### 1. Define Column ####
d <- read_csv("Cmt20_Demand.csv", col_types = "ncffnnnf",
             locale = locale(encoding = "cp874"))
d$grass_date <- ymd(d$grass_date)
d <- d %>% as_tibble(index = grass_date, key = SKU_CODE_T)
print(d)
```

```
#### 2. Train & test split ####
d_train <- d %>% filter(grass_date < ymd("2025-06-01"))
d_future <- d %>% filter(grass_date > ymd("2025-05-31"))
```

DATA EXAMPLE

```
# A tibble: 4,240 x 7 [10]
# Key:   SKU_CODE_T [20]
   Day grass_date SKU_CODE_P SKU_CODE_T item_sold Month Flag_Campaign
  <dbl> <date> <fct> <fct> <dbl> <dbl> <fct>
1 1 2025-01-01 B88B_7589354191 AAAA_1630599912 38 1 DDay
2 2 2025-01-02 B88B_7589354191 AAAA_1630599912 26 1 BAU
3 3 2025-01-03 B88B_7589354191 AAAA_1630599912 19 1 BAU
4 4 2025-01-04 B88B_7589354191 AAAA_1630599912 27 1 BAU
5 5 2025-01-05 B88B_7589354191 AAAA_1630599912 32 1 BAU
6 6 2025-01-06 B88B_7589354191 AAAA_1630599912 19 1 BAU
7 7 2025-01-07 B88B_7589354191 AAAA_1630599912 35 1 BAU
8 8 2025-01-08 B88B_7589354191 AAAA_1630599912 24 1 BAU
9 9 2025-01-09 B88B_7589354191 AAAA_1630599912 53 1 BD
10 10 2025-01-10 B88B_7589354191 AAAA_1630599912 18 1 BAU
# i 4,230 more rows
# i use `print(n = ...)` to see more rows
```



2 FORECAST MODEL DEVELOPMENT

MULTIPLE REGRESSIONS FORECAST WITH DUMMY VARIABLES

Model Formulation :

$$Y_t = \beta_0 + \beta_1 \cdot \text{Trend}_t + \sum_{i=1}^6 \gamma_i \cdot \text{Day}_i + \sum_{i=1}^4 \delta_i \cdot \text{Campaign}_t + \epsilon_t$$

Trend
Season
Flag Campaign

WHERE

Y_t = the demand (item_sold) at time t

β_0 = the intercept.

β_1 = coefficient for the linear time trend.

γ_i = coefficients for seasonal dummy variables (Thursday serving as the baseline).

δ_i = coefficient for the Campaign dummy variable (BAU serving as the baseline).

ϵ_t = the error term.

5. Model fit & Train Accuracy

```
fit <- dts_train %>%
  model(
    mod1 = TSLM(item_sold ~ trend()),
    mod2 = TSLM(item_sold ~ Flag_Campaign),
    mod3 = TSLM(item_sold ~ trend() + Flag_Campaign),
    mod4 = TSLM(item_sold ~ season()),
    mod5 = TSLM(item_sold ~ trend() + season()),
    mod6 = TSLM(item_sold ~ trend() + season() + Flag_Campaign)
  )
```



2 FORECAST MODEL DEVELOPMENT

SAMPLE SKU AAAA_1630599912CAST (MODEL 6)

Series: item_sold
Model: TSLM

Residuals:

Min	1Q	Median	3Q	Max
-51.622	-8.155	-2.082	7.577	69.585

Coefficients:

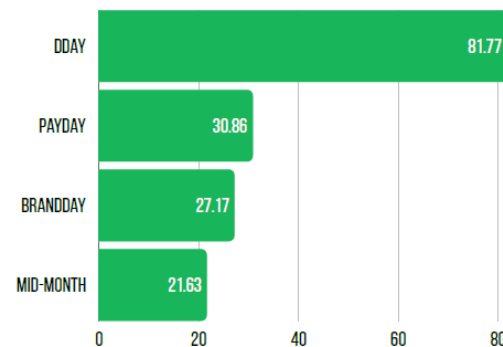
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.47182	3.54527	2.390	0.018208 *
trend()	0.20519	0.02556	8.029	3.73e-13 ***
season()week2	-1.93259	4.10437	-0.471	0.638478
season()week3	3.52991	4.11894	0.857	0.392922
season()week4	3.87085	4.17788	0.927	0.355786
season()week5	4.52728	4.19866	1.078	0.282782
season()week6	5.33100	4.18929	1.273	0.205309
season()week7	-0.82183	4.10942	-0.200	0.841783
Flag_CampaignDDay	61.76694	6.26615	13.049	< 2e-16 ***
Flag_CampaignRD	27.17249	9.88673	2.748	0.006784 **
Flag_CampaignMidMonth	21.62505	6.25424	3.458	0.000723 ***
Flag_CampaignPayday	30.86090	6.26424	4.927	2.34e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.52 on 139 degrees of freedom
Multiple R-squared: 0.6589, Adjusted R-squared: 0.6319
F-statistic: 24.41 on 11 and 139 DF, p-value: < 2.22e-16

Baseline: BAU

SKU AAAA_1630599912 COEFFICIENT



Intervention Analysis



One-period; temporary; permanent?

- Goal of intervention analysis: make adjustments to the model when the underlying structure changes
 - Competitor action
 - Government regulation
 - Supply chain disruption, e.g., worker striker, natural disaster
 - Company's advertising, e.g., an auto manufacturer's offering 0% finance to new buyers for four-week period
 - Covid lockdown dummy: **one-period, temporary, permanent?**
- One-period effect: $X_t = 1$ if $t = t_0$ and 0 otherwise (spike)
- Temporary effect: $X_t = 1$ if $t_0 \leq t \leq t_0 + \tau$ and 0 otherwise
- Permanent effect: $X_t = 1$ if $t \geq t_0$ and 0 otherwise (step function)

Interaction

Example Bank Salary: Without interaction

- Dependent variable

$Y = \text{Salary}$

- Independent variables

$X = \text{YrsExper}$

$$F = \begin{cases} 1, & \text{if female} \\ 0, & \text{otherwise} \end{cases}$$

- Regression equation

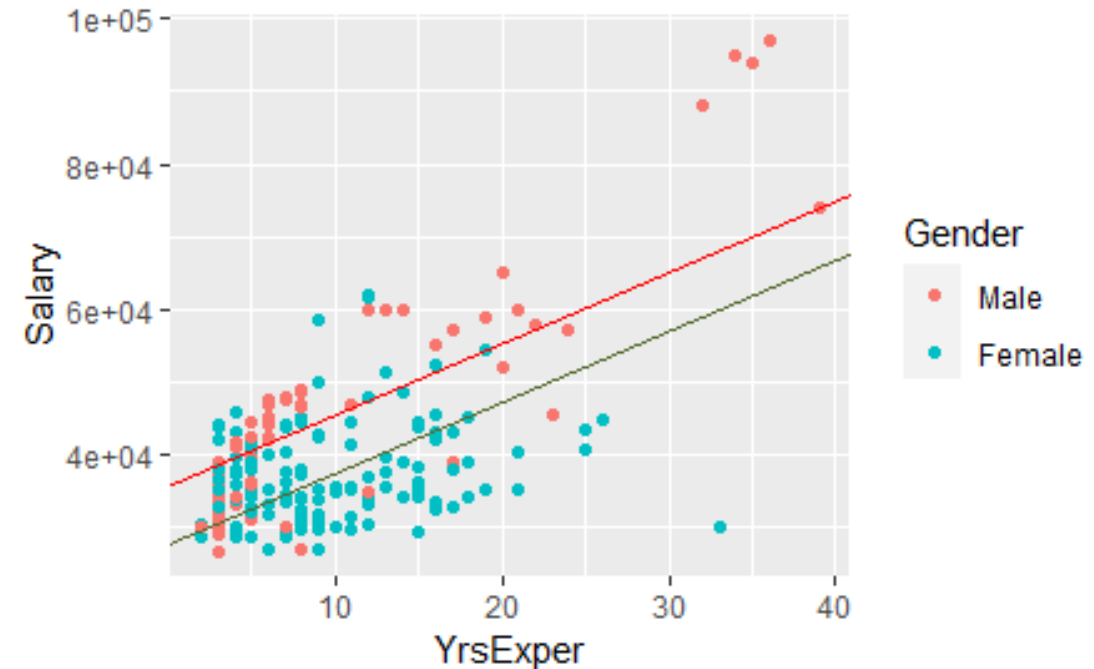
$$Y = \beta_0 + \beta_1 X + \beta_2 F = 35823.8 + 981.15X - 8011.89F$$

- Regression for male ($F = 0$)

$$Y = \beta_0 + \beta_1 X = 35823.8 + 981.2 X$$

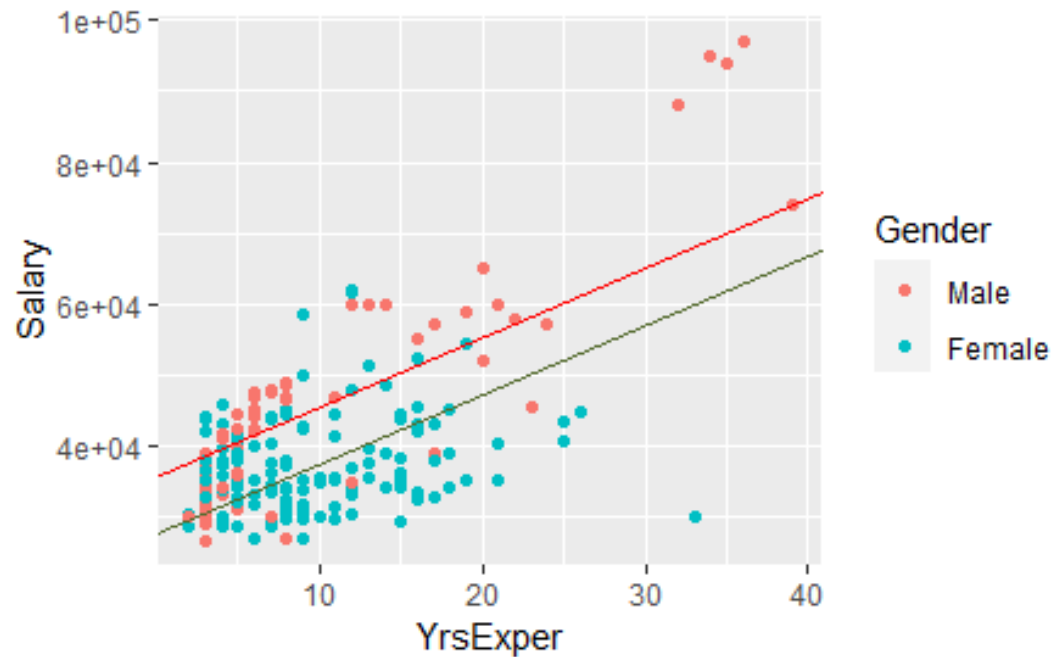
- Regression for female ($F = 1$)

$$Y = (\beta_0 + \beta_2) + \beta_1 X = 27811.9 + 981.2 X$$

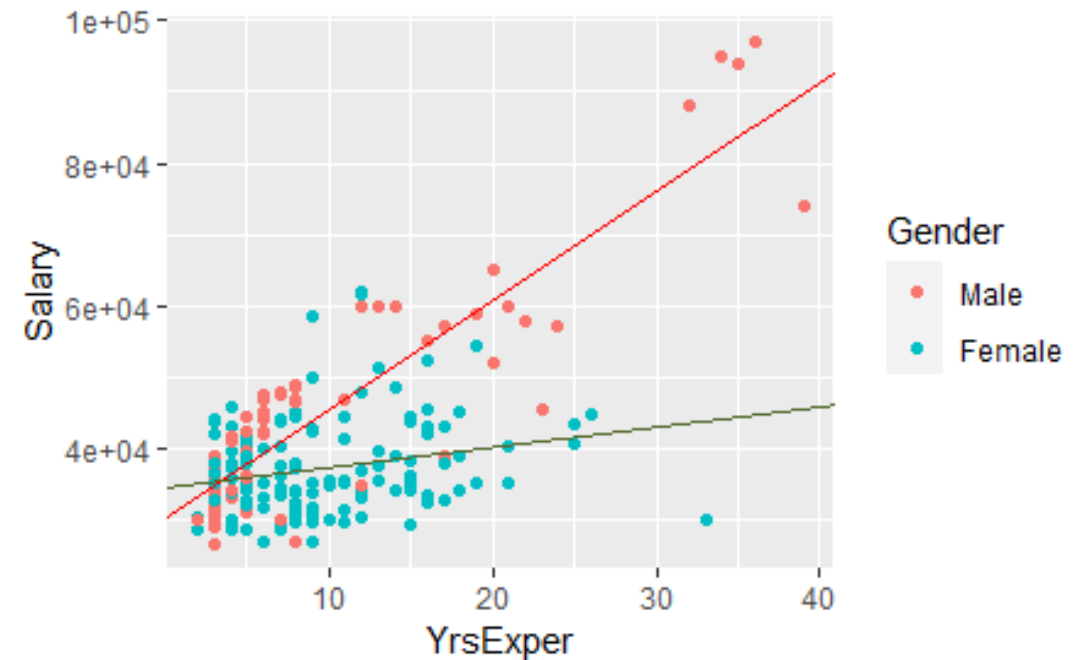


Example Bank Salary (Con.)

Without interaction



With interaction



Example (Con.): Interaction

- Dependent variable

Y = Salary

- Independent variables

X = YrsExper

$$F = \begin{cases} 1, & \text{if female} \\ 0, & \text{otherwise} \end{cases}$$

$I = XF$ **interaction variable**

- Regression equation

$$Y = \beta_0 + \beta_1 X + \beta_2 F + \beta_3 (XF)$$

$$Y = 30430 + 1528 X + 4098 F - 1248 XF$$

- Regression for male ($F = 0$)

$$Y = \beta_0 + \beta_1 X$$

$$Y = 30430 + 1528X$$

- Regression for female ($F = 1$)

$$Y = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)X$$

$$Y = 34528 + 280X$$

```
m2i <- lm(Salary ~ YrsExper*Gender, data = d)
```

```
summary(m2i)
```

R code: $a*b$ “effect of a plus effect of b plus interaction

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	30430.03	1216.57	25.01	<2e-16	***
YrsExper	1527.76	90.46	16.89	<2e-16	***
GenderFemale	4098.25	1665.84	2.46	0.0147	*
YrsExper:GenderFemale	-1247.80	136.68	-9.13	<2e-16	***

The interaction term is significant.

Intercept ($\beta_0 = 30430$)

Baseline: the predicted salary for **male** employees ($F=0$) with **0 years of experience**.

YrsExper ($\beta_1 = 1528$)

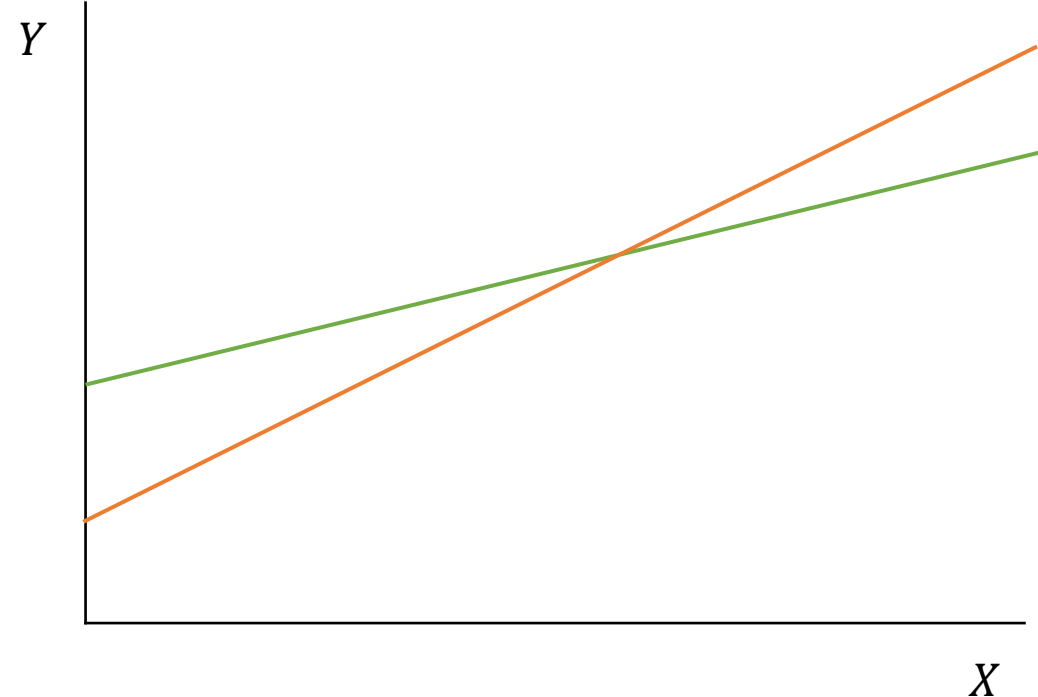
For **males**, each additional year of experience increases salary by **\$1,528** on average.

Female ($\beta_2 = 4098$)

When **YrsExper = 0**, females earn **\$4,098 more** than males (since the female line starts higher). However, this effect is modified by the interaction term as experience increases.

Interaction ($\beta_3 = -1248$)

The slope for females is **\$1,248 smaller** per year of experience than the slope for males. That is, as years of experience increase, the female advantage diminishes — eventually reversing if X is large enough.



Interactions

- Interaction between two factors.
 - Simplest case
 - Having different levels for all possible combinations of levels of the two factors.
- Interaction between a factor and a numerical variable
 - Linear effects of the continuous variable, but with different slopes within each group defined by a factor.
- Interaction between two continuous variables.
 - Slightly peculiar model.
 - Linear effect of varying one variable while keeping the other constant, but with the slope that changes as you vary the other variable.

Hw: New Car Sales

Model	R^2	Adjusted R^2	AICc (from R)
1	0.828	0.816	1132
2	0.918	0.907	1096
3			
4			
Select	N/A	Model xx	Model yy

- Model 1: $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$

$$\text{NCS.FC} = 25428.255 + 3.228\text{DPIPC} - 7666.852\text{UR} - 3219.499\text{PR} + 1122.429\text{UMICS}$$

- Model 2: $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \gamma_2 Q_2 + \gamma_3 Q_3 + \gamma_4 Q_4$

$$\text{NCS.FC} = 15351.530 + 3.197\text{DPIPC} - 7607.513\text{UR} - 3306.902\text{PR} + 1154.704\text{UMICS} + 14346.704Q_2 + 18654.821Q_3 + 1287.332Q_4$$

- Model 3

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \gamma_2 Q_2 + \gamma_3 Q_3 + \gamma_4 Q_4 + \alpha t$$

- Model 4: Allow the effect of UMICS to differ by quarter

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \gamma_2 Q_2 + \gamma_3 Q_3 + \gamma_4 Q_4 + \alpha t + \delta_2 x_4 Q_2 + \delta_3 x_4 Q_3 + \delta_4 x_4 Q_4$$

For each model,

- 1) write regression equation. For models 2, 3, 4, use baseline Q1.
- 2) Obtain R^2 , Adjusted R^2 and AICc.
- 3) For Model 4, write regression equation specifically for quarter 2.
- 4) Which model is selected based on adjusted R^2 ? Which model is selected based on AICc? (Fill in the table.)

File 07-TS-regression.R

```
#####
# Homework New Car Sales
#####

d <- read_excel("06-TS-regression.xlsx", sheet = "NewCarsSales")
glimpse(d)

ncs <- d %>%
  mutate(
    YearQuarter = yearquarter("2002 Q1") + 0:59,
    qtr = factor(quarter(YearQuarter), levels = 1:4),
    qtr = relevel(qtr, ref = "1")
  ) %>%
  select(-Date) %>%
  relocate(YearQuarter) %>%
  as_tsibble(index = YearQuarter)

ncs %>%
  model(
    mod1 = TSLM(NCS ~ DPIPC + UR + PR + UMICS),
    mod2 = TSLM(NCS ~ DPIPC + UR + PR + UMICS + season())
  ) %>%
  glance() %>%
  select(.model, r_squared, adj_r_squared, AICc)
```

.model	r_squared	adj_r_squared	AICc
<chr>	<dbl>	<dbl>	<dbl>
mod1	0.828	0.816	1132.
mod2	0.918	0.907	1096.

```
> ncs %>%
+   model(TSLM(NCS ~ DPIPC + UR + PR + UMICS + season())) %>%
+   report()
Series: NCS
Model: TSLM

Residuals:
    Min       1Q   Median       3Q      Max
-19510.1  -5010.2   152.8   6146.3  17196.4

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 15351.5297  26841.1403   0.572  0.56983
DPIPC         3.1970    0.2791  11.456 7.75e-16 ***
UR        -7607.5129  1186.9485  -6.409 4.32e-08 ***
PR        -3306.9021  1005.4594  -3.289 0.00181 **
UMICS       1154.2495   150.9923   7.644 4.66e-10 ***
season()year2 14346.7036  3039.8505   4.720 1.82e-05 ***
season()year3 18654.8207  3055.9747   6.104 1.32e-07 ***
season()year4  1287.3318   3058.0208   0.421 0.67551
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8319 on 52 degrees of freedom
Multiple R-squared: 0.9177,    Adjusted R-squared: 0.90664
F-statistic: 82.85 on 7 and 52 DF, p-value: < 2.22e-16
```

Appendix

Full Rank Assumption: Dummy Variable Trap

- Recall that multiple regression can be written as

$$y_t = \beta_0 + \beta_1 x_{1,t} + \dots + \beta_k x_{k,t} + \epsilon_t$$

- Let $\mathbf{y} = (y_1, \dots, y_T)^\top$, $\boldsymbol{\beta} = (\beta_0, \dots, \beta_k)^\top$, $\boldsymbol{\epsilon} = (\epsilon_1, \dots, \epsilon_T)^\top$,

- Matrix formulation: $\mathbf{y} = \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\epsilon}$

- Least squares estimation: Minimize $\boldsymbol{\epsilon}^\top \boldsymbol{\epsilon} = (\mathbf{y} - \mathbf{X} \boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X} \boldsymbol{\beta})$

$$\mathbf{X} = \begin{bmatrix} 1 & x_{1,1} & x_{2,1} & \dots & x_{k,1} \\ 1 & x_{1,2} & x_{2,2} & \dots & x_{k,2} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{1,T} & x_{2,T} & \dots & x_{k,T} \end{bmatrix}$$

- The least squares estimate $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$
- The estimated coefficients require the inversion of matrix $\mathbf{X}^\top \mathbf{X}$.
- Rank deficiency:** If X is not of full column rank, then matrix $\mathbf{X}^\top \mathbf{X}$ is singular and the model cannot be estimated.
- This will occur, for example, if you fall for the “dummy variable trap”, i.e., having the same number of dummy variables as there are categories of a categorical predictor.
 - Multicollinearity

Dummy Variable Trap Example

Event indices	Description
1	Nothing
2	Trade promotion
3	Media (e.g., radio, TV) ad campaign

Event	Description	E1
1	Nothing	1
2	Promotion	0
3	Media	0

E2	E3
0	0
1	0
0	1

$X =$

Ones	Time	E1	E2	E3
1	1	0	0	1
1	2	0	0	1
1	3	0	1	0
1	4	1	0	0
1	5	1	0	0
1	6	1	0	0
1	7	1	0	0
1	8	1	0	0
1	9	0	1	0
1	10	1	0	0
1	11	1	0	0
1	12	1	0	0
1	13	0	1	0
1	14	1	0	0
1	15	0	1	0
1	16	1	0	0
1	17	1	0	0
1	18	1	0	0
1	19	1	0	0
1	20	1	0	0
1	21	0	1	0
1	22	1	0	0
1	23	1	0	0
1	24	1	0	0
1	25	0	1	0
1	26	1	0	0
1	27	0	1	0

- $E_1 + E_2 + E_3 = 1$
- E_1 is a linear combination of E_2 and E_3 .
- The matrix $X^T X$ is not full column rank.
- The number of columns is 5.
- $\text{Rank}(X) = 4$.
- Not full rank.

```
> qr(X)$rank # rank of matrix
```

```
[1] 4
```

```
> t(X)%*%X
```

```
      [,1] [,2] [,3] [,4] [,5]
[1,]  27  378  18   7   2
[2,]  378 6930 262 113   3
[3,]   18  262  18   0   0
[4,]    7  113   0   7   0
[5,]    2    3   0   0   2
```

```
> det(t(X)%*%X) # determinant of matrix
```

```
[1] -8.64711e-10
```

```
> solve(t(X)%*%X) # inverse of matrix
```

```
Error in solve.default(t(X) %*% X) :
system is computationally singular: rec
```

One-Hot Encoding vs Dummy variables

Feature	One-Hot Encoding	Dummy Variables
Origin	Machine Learning (Computer Science)	Statistics (Econometrics)
New Columns	N categories = N columns	N categories = $N - 1$ columns
Reference Cat.	No reference category (includes all)	Uses a reference category (drops one)
Multicollinearity	High risk (Dummy Variable Trap)	Low risk (avoids Trap)
Best For	Neural Networks, Tree-based models	Linear Regression, Logistic Regression